THE UNIVERSITY OF BAMENDA

NATIONAL HIGHER POLYTECHNIC INSTITUTE

ENGINEERING ECONOMICS (PART I: THEORY)
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Chapter 1: Introduction to Engineering Economics

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CHAPTER 1: Introduction to Engineering Economics

1.1 Introduction

- **Engineering economics** is concerned with the formulation, estimation and evaluation of the economic outcomes of alternatives that are available to accomplish a defined purpose. **Engineering economics** can be defined as a collection of mathematical techniques that simplify economic comparison. Engineers use the knowledge of engineering economics in analyzing, synthesizing and drawing conclusions as they work on projects of different sizes. In other words, the techniques and models of engineering economics assist engineers in making decisions. The success of engineering and business projects is normally measured in terms of financial efficiency. A project will be able to achieve maximum financial efficiency when it is properly planned and operated with respect to its technical, social, and financial requirements. Since engineers understand the technical requirements of a project, they can combine the technical details of the project and the knowledge of engineering economics to study and arrive at a sound managerial decision. The basic economic concepts that are essentially factors to be considered in making economy studies are briefly discussed in this chapter
- **Economics** is the science that deals with the production and consumption of goods and services and the distribution and rendering of these for human

welfare.

The following are the economic goals.

- ✓ A high level of employment
- ✓ Price stability
- ✓ Efficiency
- ✓ An equitable distribution of income
- ✓ Growth

- **Scarcity**: the excess of human needs over what can actually be produced."
  - ✓ Economics is the study of choices in the face of scarcity of resources.

Why is scarcity a problem?

- ✓ If we know that we have limited resources we could just behave accordingly. The problem arises because human wants and needs are virtually unlimited but resources available to satisfy them are not.

- ✓ Scarcity implies that we face some sort of constraints every time we take an economic decision.
- ✓ The presence of constraints has the main implication of creating trade-offs among different alternatives. The concept of trade-off is one of the core principles in economics.
- ✓ Economics is the study of choices under conditions of scarcity, or the study of choice with constraints

- **Trade-off**: The idea that because of scarcity, producing more of one good or service means producing less of another good or service. Trade-offs force society to make choices.
  - ✓ Trade-offs forces engineers to make choices, particularly when answering the following fundamental questions:
    – Which engineering projects are worthwhile?
    – Which engineering projects should have a higher priority?
    – How should the engineering project be designed?
  - ✓ As summarized by the concept of trade-off, any choice, made when resources are scarce, involves some sacrifice.
  - ✓ The opportunity cost expresses "the basic relationship between scarcity and choice"

1.2 Flow in an Economy

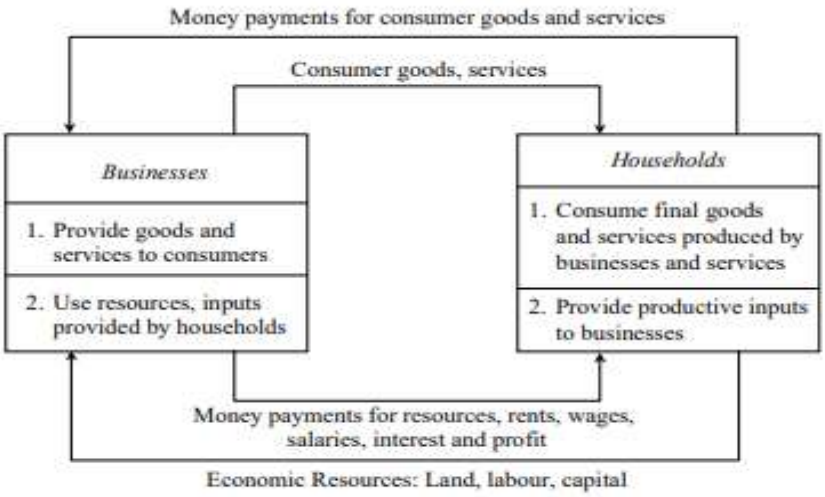


Fig. 1.1 Flow of goods, services, resources and money payments in a simple economy.

1.3 Law of Supply and Demand

An interesting aspect of the economy is that the demand and supply of a product are interdependent and they are sensitive with respect to the price of that product. The interrelationships between them are shown in Fig. below. **It is clear that when there is a decrease in the price of a product, the demand for the product increases and its supply decreases. Also, the product is more in demand and hence the demand of the product increases. At the same time, lowering of the price of the product makes the producers restrain from releasing more quantities of the product in the market. Hence, the supply of the product is decreased**. The point of intersection of the supply curve and the demand curve is known as the equilibrium point. At the price corresponding to this point, the quantity of supply is equal to the quantity of demand. Hence, this point is called the equilibrium point.

1.3.1 Factors influencing demand

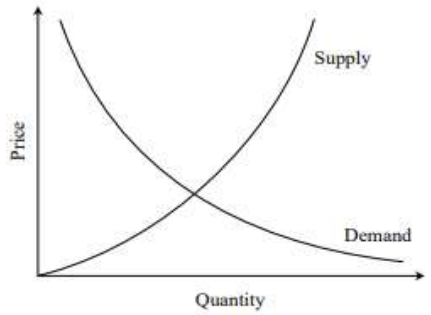
The shape of the demand curve is influenced by the following factors:

- ✓ Income of the people
- ✓ Prices of related goods
- ✓ Tastes of consumers

1.3.2 Factors influencing supply

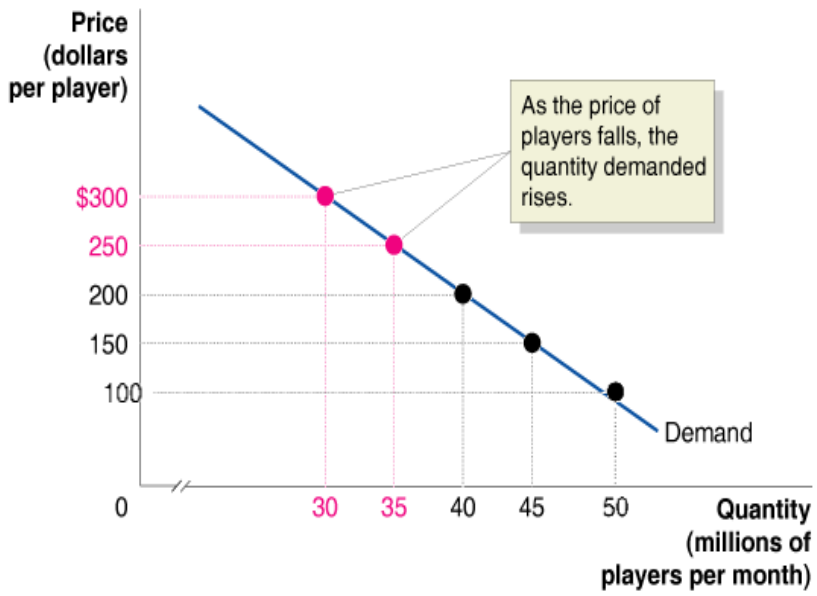
The shape of the supply curve is affected by the following factors:

- ✓ Cost of the inputs
- ✓ Technology
- ✓ Weather
- ✓ Prices of related goods

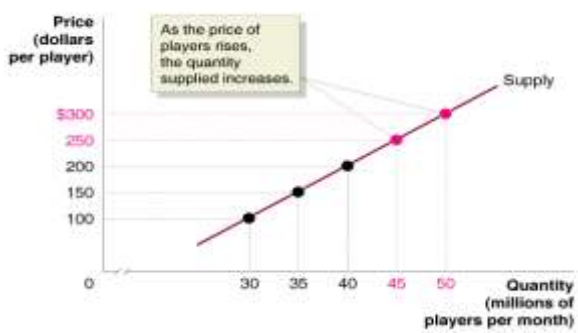


Demand and supply curve

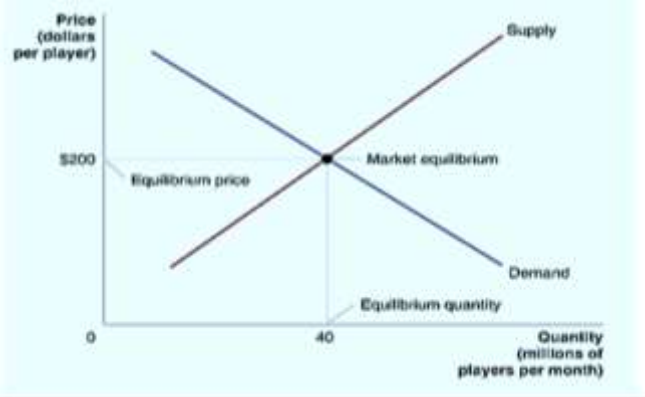
- ✓ **Law of demand**: The rule that, holding everything else constant, when the price of a product falls, the quantity demanded of the product will increase, and when the price of a product rises, the quantity demanded of the product will decrease.



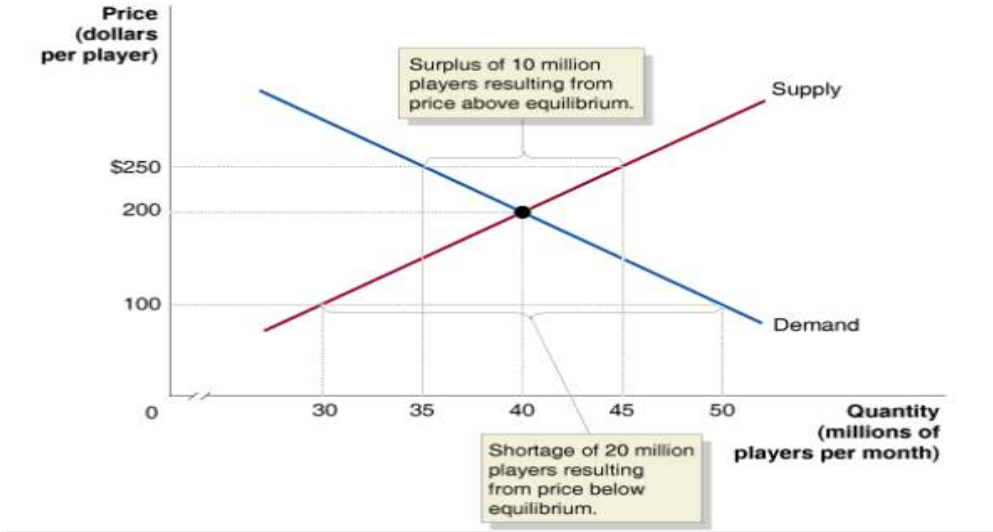
- ✓ **Law of supply:** The rule that, holding everything else constant, increases in price cause increases in the quantity supplied, and decreases in price cause decreases in the quantity supplied.



- ✓ **Market equilibrium**: A situation in which quantity demanded equals quantity supplied



- ✓ **Surplus**: A situation in which the quantity supplied is greater than the quantity demanded
- ✓ **Shortage:** A situation in which the quantity demanded is greater than the quantity supplied



1.4 Concept of engineering economics

Science is a field of study where the basic principles of different physical systems are formulated and tested. Engineering is the application of science. It establishes varied application systems based on different scientific principles. From the discussions in the previous section, it is clear that price has a major role in deciding the demand and supply of a product. Hence, from the organization's point of view, efficient and effective functioning of the organization would certainly help it to provide goods/services at a lower cost which in turn will enable it to fix a lower price for its goods or services. The following section discusses the

different types of efficiency and their impact on the operation of businesses and the definition and scope of engineering economics

- **Types of Efficiency**

Efficiency of a system is generally defined as the ratio of its output to input. The efficiency can be classified into technical efficiency and economic efficiency.

Technical efficiency It is the ratio of the output to input of a physical system. The physical system may be a diesel engine, a machine working in a shop floor, a furnace, etc.

$$\text{Technical efficiency (\%)} = \frac{\text{Output}}{\text{Input}} \times 100$$

The technical efficiency of a diesel engine is as follows:

$$\text{Technical efficiency (\%)} = \frac{\text{Heat equivalent of mechanical energy produced}}{\text{Heat equivalent of fuel used}} \times 100$$

In practice, technical efficiency can never be more than 100%. This is mainly due to frictional loss and incomplete combustion of fuel, which are considered to be unavoidable phenomena in the working of a diesel engine.

Economic efficiency is the ratio of output to input of a business system

$$\text{Economic efficiency (\%)} = \frac{\text{Output}}{\text{Input}} \times 100 = \frac{\text{Worth}}{\text{Cost}} \times 100$$

'**Worth**' is the annual revenue generated by way of operating the business and '**cost**' is the total annual expenses incurred in carrying out the business. For the survival and growth of any business, the economic efficiency should be more than 100%. Economic efficiency is also called 'productivity'. There are several ways of improving productivity.

- ✓ Increased output for the same input
- ✓ Decreased input for the same output
- ✓ By a proportionate increase in the output which is more than the proportionate increase in the input

- ✓ By a proportionate decrease in the input which is more than the proportionate decrease in the output
- ✓ Through simultaneous increase in the output with decrease in the input

Group work

CHAPTER 2: Elementary Economic Analysis

2.1 Overview

Whether it is a business situation or a day-to-day event in somebody's personal life, there are a large number of economic decisions making involved. One can manage many of these decision problems by using simple economic analysis. For example, an industry can source its raw materials from a nearby place or from a far-off place. In this problem, the following factors will affect the decision:

- ✓ Price of the raw material
- ✓ Transportation cost of the raw material
- ✓ Availability of the raw material
- ✓ Quality of the raw material

Consider the alternative of sourcing raw materials from a nearby place with the following characteristics:

- ✓ The raw material is more costly in the nearby area.
- ✓ The availability of the raw material is not sufficient enough to support the operation of the industry throughout the year.
- ✓ The raw material requires pre-processing before it is used in the production process. This would certainly add cost to the product.
- ✓ The cost of transportation is minimal under this alternative.

On the other hand, consider another alternative of sourcing the raw materials from a far-off place with the following characteristics:

- ✓ The raw material is less costly at the far-off place.
- ✓ The cost of transportation is very high.
- ✓ The availability of the raw material at this site is abundant and it can support the plant throughout the year.
- ✓ The raw material from this site does not require any pre-processing before using it for production.

Under such a situation, the procurement of the raw material should be decided in such a way that the overall cost is minimized

The above example clearly highlights the various components of cost that are involved in each of the alternatives of the decision-making process as well as a method of taking a suitable decision

2.2 Examples for simple economic analysis

In this section, the concept of simple economic analysis is illustrated using suitable examples in the following areas:

- ✓ Material selection for a product
- ✓ Design selection for a product
- ✓ Design selection for a process industry
- ✓ Building material selection for construction activities
- ✓ Process planning/Process modification

2.2.1 Material Selection for a Product/Substitution of Raw Material

The cost of a product can be reduced greatly by substitution of the raw materials. Among various elements of cost, raw material cost is most significant and it forms a major portion of the total cost of any product. So, any attempt to find a suitable raw material will bring a reduction in the total cost in any one or combinations of the following ways:

- ✓ Cheaper raw material price
- ✓ Reduced machining/process time
- ✓ Enhanced durability of the product

Therefore, the process of raw material selection/substitution will result in finding an alternate raw material which will provide the necessary functions that are provided by the raw material that is presently used. In this process, if the new raw material provides any additional benefit, then it should be treated as its welcoming feature. This concept is demonstrated with two numerical problems.

EXAMPLE 1

2.2.2 Design Selection for a Product

The design modification of a product may result in reduced raw material requirements, increased machinability of the materials and reduced labor. Design is an important factor which decides the cost of the product for a specified level of performance of that product. The elementary economic analysis applied to the selection of design for a product is illustrated with two example problems.

EXAMPLE 2

2.2.3 Building Material Selection

As discussed in the introduction to this chapter, the sourcing of raw materials will have a significant effect on the cost of any product. Hence, it is assumed that the price of raw material is location dependent. While sourcing a raw material, the cost of transportation is to be considered in conjunction with the price of the raw material. This concept is demonstrated with a numerical example.

EXAMPLE 3

2.2.4 Process Planning /Process Modification

While planning for a new component, a feasible sequence of operations with the least cost of processing is to be considered. The process sequence of a component which has been planned in the past is not static. It is always subject to modification with a view to minimize the cost of manufacturing the component. So, the objective of process planning/process modification is to identify the most economical sequence of operations to produce a component.

The steps in process planning are as follows:

1. Analyze the part drawing to get an overall picture of what is required.
2. Make recommendations to or consult with product engineers on product design changes.
3. List the basic operations required to produce the part to the drawing or specifications.
4. Determine the most practical and economical manufacturing method and the form or tooling required for each operation.
5. Devise the best way to combine the operations and put them in sequence.
6. Specify the gauging required for the process.

Steps 3–5 aim to determine the most practical and economical sequence of operations to produce a component. This concept is demonstrated with a numerical problem.

EXAMPLE 4

2.3 Fundamental principles of engineering economics

Principle 1: A nearby dollar is worth more than a distant dollar

Principle 2: All it counts is the differences among alternatives

Principle 3: Marginal revenue must exceed marginal cost

Principle 4: Additional risk is not taken without the expected additional return

CHAPTER 3: Interest Formulas and Their Applications

Interest rate is the rental value of money. It represents the growth of capital per unit period. The period may be a month, a quarter, semiannual or a year. An interest rate 15% compounded annually means that for every hundred francs invested now, an amount of 15 frs will be added to the account at the end of the first year. So, the total amount at the end of the first year will be 115 frs. At the end of the second year, again 15% of 115 frs, i.e. 17.25frs will be added to the account. Hence the total amount at the end of the second year will be 132.25 frs. The process will continue thus till the specified number of years.

1.1 Time value of money

If an investor invests a sum of 100frs in a fixed deposit for five years with an interest rate of 15% compounded annually, the accumulated amount at the end of every year will be as shown in Table

(amount of deposit = Rs. 100.00)

Year end	*Interest* (Rs.)	*Compound amount* (Rs.)
0		100.00
1	15.00	115.00
2	17.25	132.25
3	19.84	152.09
4	22.81	174.90
5	26.24	201.14

The formula to find the future worth in the third column is

$$F = P \times (1 + i)^n$$

where

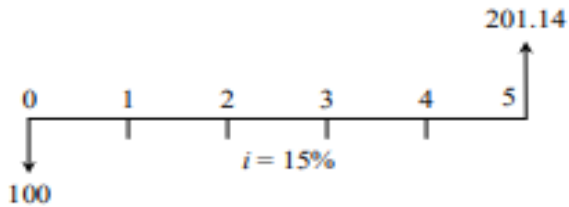
P = principal amount invested at time 0,

F = future amount,

i = interest rate compounded annually,

n = period of deposit.

The maturity value at the end of the fifth year is 201.14 frs. This means that the amount 201.14 frs at the end of the fifth year is equivalent to 100.00 frs at time 0 (i.e. at present). This is diagrammatically shown in Fig.. This explanation assumes that the inflation is at zero percentage.



Time value of money

Alternatively, the above concept may be discussed as follows: If we want 100.00 frs at the end of the nth year, what is the amount that we should deposit now at a given interest rate, say 15%? A detailed working is shown in Table below

(rate of interest = 15%)

End of year (n)	*Present worth*	*Compound amount after n year(s)*
0		100
1	86.96	100
2	75.61	100
3	65.75	100
4	57.18	100
5	49.72	100
6	43.29	100
7	37.59	100
8	32.69	100
9	28.43	100
10	24.72	100

The formula to find the present worth in the second column is

$$P = \frac{F}{(1+i)^n}$$

From Table, it is clear that if we want 100 frs at the end of the fifth year, we should now deposit an amount of 49.72 frs. Similarly, if we want 100.00 frs at the end of the 10th year, we should now deposit an amount of 24.72 frs.

1.2 Interest formulas

While making investment decisions, computations will be done in many ways. To simplify all these computations, it is extremely important to know how to use interest formulas more effectively. Before discussing the effective application of the interest formulas for investment-decision making, the various interest formulas are presented first.

Interest rate can be classified into **simple interest rate** and **compound interest rate**.

In simple interest, the interest is calculated, based on the initial deposit for every interest period. In this case, calculation of interest on interest is not applicable.

In compound interest, the interest for the current period is computed based on the amount (principal plus interest up to the end of the previous period) at the beginning of the current period. The notations which are used in various interest formulae are as follows:

P = principal amount

n = No. of interest periods

i = interest rate (It may be compounded monthly, quarterly, semiannually or annually)

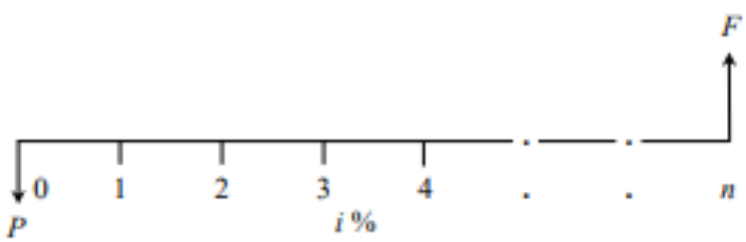
F = future amount at the end of year n

A = equal amount deposited at the end of every interest period

G = uniform amount which will be added/subtracted period after period to/ from the amount of deposit A1 at the end of period

1.2.1 Single-Payment Compound Amount

Here, the objective is to find the single future sum (F) of the initial payment (P) made at time 0 after n periods at an interest rate i compounded every period. The cash flow diagram of this situation is shown in Fig.



The formula to obtain the single-payment compound amount is

$$F = P(1 + i)^n = P(F/P, i, n)$$

where

(F/P, i, n) is called as single-payment compound amount factor

EXAMPLE 5

3.2.2 Single-Payment Present Worth Amount

Here, the objective is to find the present worth amount (P) of a single future sum (F) which will be received after n periods at an interest rate of i compounded at the end of every interest period. The formula to obtain the present worth is

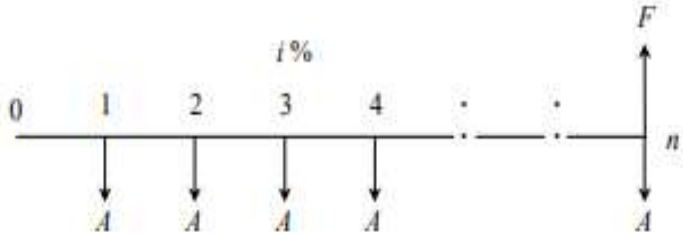
$$P = \frac{F}{(1+i)^n} = F(P/F, i, n)$$

where (P/F, i, n) is termed as single-payment present worth factor

EXAMPLE 6

1.2.2 Equal-Payment Series Compound Amount

In this type of investment mode, the objective is to find the future worth of n equal payments which are made at the end of every interest period till the end of the nth interest period at an interest rate of i compounded at the end of each interest period. The corresponding cash flow diagram is shown in Fig.



Cash flow diagram of equal-payment series compound amount.

A = equal amount deposited at the end of each interest period

n = No. of interest periods

i = rate of interest

F = single future amount

The formula to get F is

$$F = A\,\frac{(1+i)^n - 1}{i} = A(F/A, i, n)$$

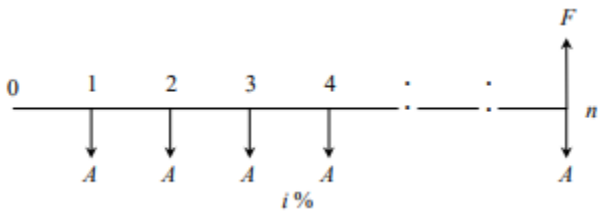
where

(F/A, i, n) is termed as equal-payment series compound amount factor

EXAMPLE 7

3.3.4 Equal-Payment Series Sinking Fund

In this type of investment mode, the objective is to find the equivalent amount (A) that should be deposited at the end of every interest period for n interest periods to realize a future sum (F) at the end of the nth interest period at an interest rate of i. The corresponding cash flow diagram is shown in Fig.



Cash flow diagram of equal-payment series sinking fund

A = equal amount to be deposited at the end of each interest period

n = No. of interest periods

i = rate of interest

F = single future amount at the end of the nth period

The formula to get F is

$$A = F\frac{i}{(1+i)^n - 1} = F(A/F, i, n)$$

EXAMPLE 8

3.3.5 Equal-Payment Series Present Worth Amount

The objective of this mode of investment is to find the present worth of an equal payment made at the end of every interest period for n interest periods at an

interest rate of i compounded at the end of every interest period.

The corresponding cash flow diagram is shown in Fig. 3.8. Here,

P = present worth

A = annual equivalent payment

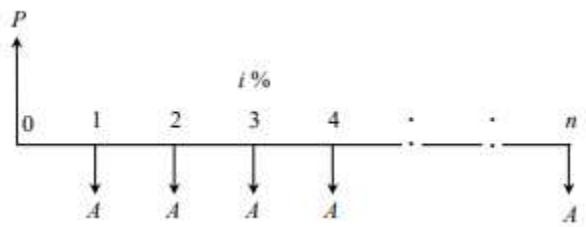
i = interest rate

n = No. of interest periods

The formula to compute P is

$$P = A\,\frac{(1+i)^n - 1}{i(1+i)^n} = A(P/A,\ i,\ n)$$

where (P/A, i, n) is called equal-payment series present worth factor

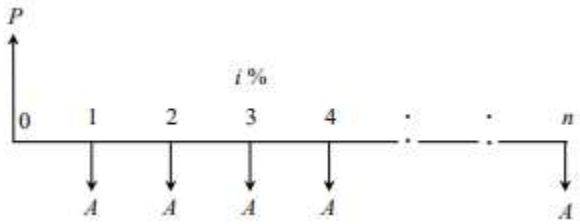


Cash flow diagram of equal-payment series present worth amount.

EXAMPLE 9

3.3.6 Equal-Payment Series Capital Recovery Amount

The objective of this mode of investment is to find the annual equivalent amount (A) which is to be recovered at the end of every interest period for n interest periods for a loan (P) which is sanctioned now at an interest rate of i compounded at the end of every interest period



Cash flow diagram of equal-payment series capital recovery amount.

P = present worth (loan amount)

A = annual equivalent payment (recovery amount)

i = interest rate

n = No. of interest periods

The formula to compute P is as follows:

$$A = P\frac{i(1+i)^n}{(1+i)^n - 1} = P(A/P, i, n)$$

where, (A/P, i, n) is called equal-payment series capital recovery factor.

EXAMPLE 10

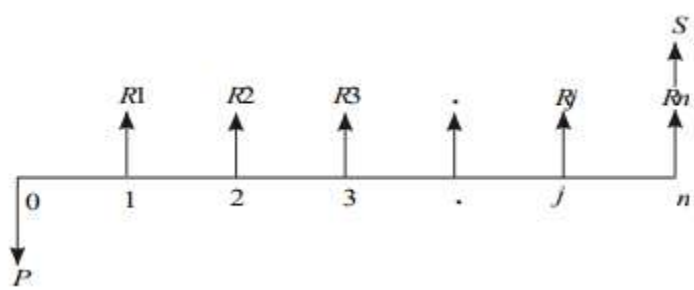
Chapter 4: Present Worth Method of Comparison, Future Worth Method, Annual Equivalent Method and Rate of Return Method

4.1 Present Worth Method of Comparison

In this method of comparison, the cash flows of each alternative will be reduced to time zero by assuming an interest rate i. Then, depending on the type of decision, the best alternative will be selected by comparing the present worth amounts of the alternatives. The sign of various amounts at different points in time in a cash flow diagram is to be decided based on the type of the decision problem. In a cost dominated cash flow diagram, the costs (outflows) will be assigned with positive sign and the profit, revenue, salvage value (all inflows), etc. will be assigned with negative sign. In a revenue/profit-dominated cash flow diagram, the profit, revenue, salvage value (all inflows to an organization) will be assigned with positive sign. The costs (outflows) will be assigned with negative sign. In case the decision is to select the alternative with the minimum cost, then the alternative with the least present worth amount will be selected. On the other hand, if the decision is to select the alternative with the maximum profit, then the alternative with the maximum present worth will be selected.

4.1.1 **Revenue-dominated cash flow diagram**

A generalized revenue-dominated cash flow diagram to demonstrate the present worth method of comparison is presented in Fig.



Revenue-dominated cash flow diagram.

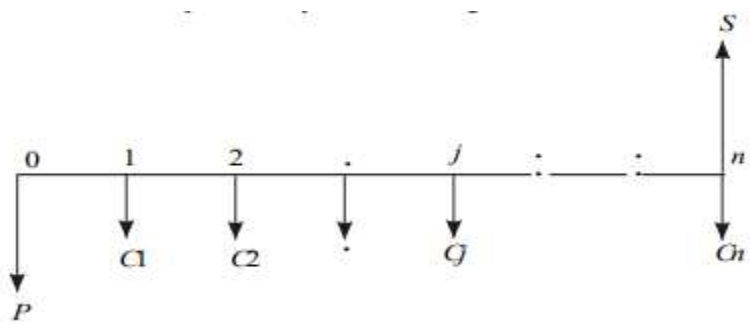
In Fig., P represents an initial investment and Rj the net revenue at the end of the jth year. The interest rate is i, compounded annually. S is the salvage value at the end of the nth year. To find the present worth of the above cash flow diagram for a given interest rate, the formula is

$$PW(i) = -P + R1[1/(1+i)^1] + R2[1/(1+i)^2] + \ldots$$
$$+ Rj[1/(1+i)^j] + Rn[1/(1+i)^n] + S[1/(1+i)^n]$$

In this formula, expenditure is assigned a negative sign and revenues are assigned a positive sign. If we have some more alternatives which are to be compared with this alternative, then the corresponding present worth amounts are to be computed and compared. Finally, the alternative with the maximum present worth amount should be selected as the best alternative.

4.1.2 Cost-dominated cash flow diagram

A generalized cost-dominated cash flow diagram to demonstrate the present worth method of comparison is presented in Fig.



In Fig., P represents an initial investment, Cj the net cost of operation and maintenance at the end of the jth year, and S the salvage value at the end of the nth year. To compute the present worth, amount of the above cash flow diagram for a given interest rate i, we have the formula

$$PW(i) = P + C1[1/(1+i)^1] + C2[1/(1+i)^2] + \ldots + Cj[1/(1+i)^j] + Cn[1/(1+i)^n] - S[1/(1+i)^n]$$

In the above formula, the expenditure is assigned a positive sign and the revenue a negative sign. If we have some more alternatives which are to be compared with this alternative, then the corresponding present worth amounts are to be computed and compared. Finally, the alternative with the minimum present worth amount should be selected as the best alternative.

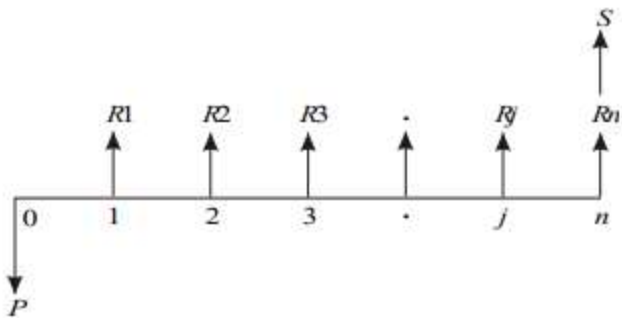
EXAMPLE 10

4.2 Future worth method

In the future worth method of comparison of alternatives, the future worth of various alternatives will be computed. Then, the alternative with the maximum future worth of net revenue or with the minimum future worth of net cost will be selected as the best alternative for implementation.

4.2.1 Revenue-dominated cash flow diagram

A generalized revenue-dominated cash flow diagram to demonstrate the future worth method of comparison is presented in Fig.



Revenue-dominated cash flow diagram

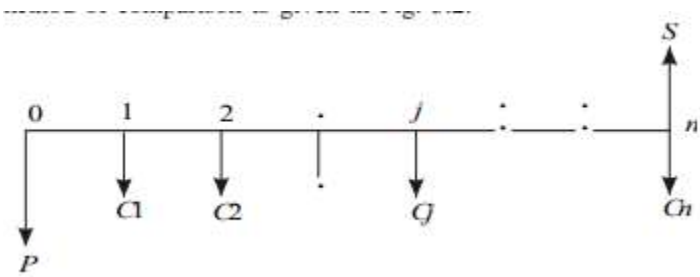
In Fig., P represents an initial investment, Rj the net-revenue at the end of the jth year, and S the salvage value at the end of the nth year. The formula for the future worth of the above cash flow diagram for a given interest rate, i is

$$FW(i) = -P(1+i)^n + R1(1+i)^{n-1} + R2(1+i)^{n-2} + \ldots + Rj(1+i)^{n-j} + \ldots + Rn + S$$

In the above formula, the expenditure is assigned with negative sign and the revenues are assigned with positive sign. If we have some more alternatives which are to be compared with this alternative, then the corresponding future worth amounts are to be computed and compared. Finally, the alternative with the maximum future worth amount should be selected as the best alternative

4.2.2 Cost-dominated cash flow diagram

A generalized cost-dominated cash flow diagram to demonstrate the future worth method of comparison is given in Fig.



In Fig., P represents an initial investment, Cj the net cost of operation and maintenance at the end of the jth year, and S the salvage value at the end of the nth year. The formula for the future worth of the above cash flow diagram for a given interest rate, i is

$$FW(i) = P(1+i)^n + C1(1+i)^{n-1} + C2(1+i)^{n-2} + \ldots + Cj(1+i)^{n-j} + \ldots + Cn - S$$

In this formula, the expenditures are assigned with positive sign and revenues with negative sign. If we have some more alternatives which are to be compared with this alternative, then the corresponding future worth amounts are to be computed and compared. Finally, the alternative with the minimum future worth amount should be selected as the best alternative

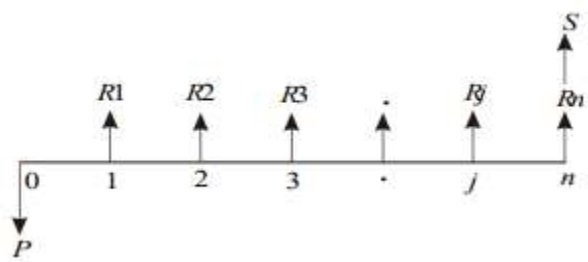
EXAMPLE 11

4.3 Annual equivalent method

In the annual equivalent method of comparison, first the annual equivalent cost or the revenue of each alternative will be computed. Then the alternative with the maximum annual equivalent revenue in the case of revenue-based comparison or with the minimum annual equivalent cost in the case of cost-based comparison will be selected as the best alternative.

4.3.2 Revenue-dominated cash flow diagram

A generalized revenue-dominated cash flow diagram to demonstrate the annual equivalent method of comparison is presented in Fig.



Revenue-dominated cash flow diagram.

In Fig, P represents an initial investment, Rj the net revenue at the end of the jth year, and S the salvage value at the end of the nth year. The first step is to find the net present worth of the cash flow diagram using the following expression for a given interest rate, i:

$$PW(i) = -P + R1/(1+i)^1 + R2/(1+i)^2 + \ldots + Rj/(1+i)^j + \ldots + Rn/(1+i)^n + S/(1+i)^n$$

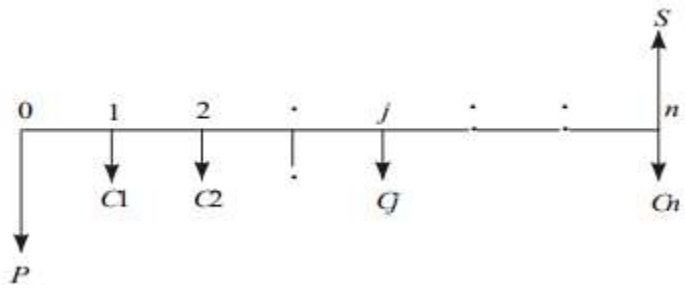
In the above formula, the expenditure is assigned with a negative sign and the revenues are assigned with a positive sign. In the second step, the annual equivalent revenue is computed using the following formula:

$$A = PW(i)\,\frac{i(1+i)^n}{(1+i)^n - 1} = PW(i)\,(A/P, i, n)$$

where (A/P, i, n) is called equal payment series capital recovery factor. If we have some more alternatives which are to be compared with this alternative, then the corresponding annual equivalent revenues are to be computed and compared. Finally, the alternative with the maximum annual equivalent revenue should be selected as the best alternative.

6.3 Cost-dominated cash flow diagram

A generalized cost-dominated cash flow diagram to demonstrate the annual equivalent method of comparison is illustrated in Fig.



Cost-dominated cash flow diagram

In Fig, P represents an initial investment, Cj the net cost of operation and maintenance at the end of the jth year, and S the salvage value at the end of the nth year. The first step is to find the net present worth of the cash flow diagram using the following relation for a given interest rate, i.

$$PW(i) = P + C1/(1+i)^1 + C2/(1+i)^2 + \ldots + Cj/(1+i)^j + \ldots + Cn/(1+i)^n - S/(1+i)^n$$

In the above formula, each expenditure is assigned with positive sign and the salvage value with negative sign. Then, in the second step, the annual equivalent cost is computed using the following equation:

$$A = PW(i)\,\frac{i(1+i)^n}{(1+i)^n - 1} = PW(i)\,(A/P, i, n)$$

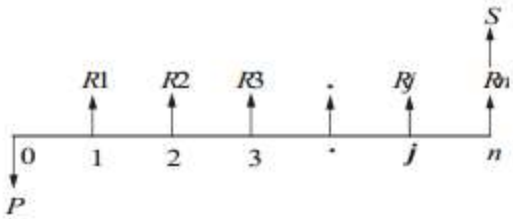
where (A/P, i, n) is called as equal-payment series capital recovery factor.

As in the previous case, if we have some more alternatives which are to be compared with this alternative, then the corresponding annual equivalent costs are to be computed and compared. Finally, the alternative with the minimum annual equivalent cost should be selected as the best alternative. If we have some non-standard cash flow diagram, then we will have to follow the general procedure for converting each and every transaction to time zero and then convert the net present worth into an annual equivalent cost/ revenue depending on the type of the cash flow diagram. Such procedure is to be applied to all the alternatives and finally, the best alternative is to be selected.

EXAMPLE 12

4.4 Rate of return method

The rate of return of a cash flow pattern is the interest rate at which the present worth of that cash flow pattern reduces to zero. In this method of comparison, the rate of return for each alternative is computed. Then the alternative which has the highest rate of return is selected as the best alternative. In this type of analysis, the expenditures are always assigned with a negative sign and the revenues/inflows are assigned with a positive sign. A generalized cash flow diagram to demonstrate the rate of return method of comparison is presented in Fig

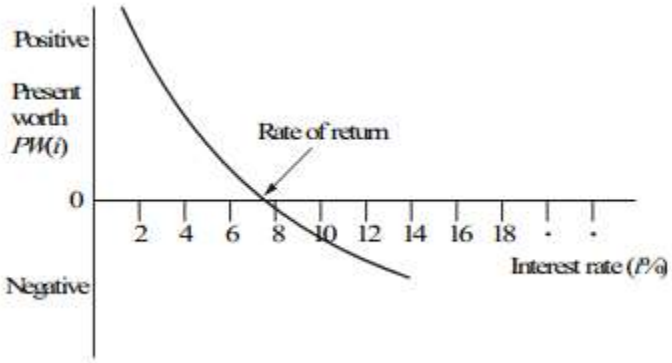


Generalized cash flow diagram

In the above cash flow diagram, P represents an initial investment, Rj the net revenue at the end of the jth year, and S the salvage value at the end of the nth year. The first step is to find the net present worth of the cash flow diagram using the following expression at a given interest rate, i

$$PW(i) = -P + R1/(1+i)^1 + R2/(1+i)^2 + \ldots + Rj/(1+i)^j + \ldots + Rn/(1+i)^n + S/(1+i)^n$$

Now, the above function is to be evaluated for different values of i until the present worth function reduces to zero, as shown in Fig.. In the figure, the present worth goes on decreasing when the interest rate is increased. The value of i at which the present worth curve cuts the X-axis is the rate of return of the given proposal/project. It will be very difficult to find the exact value of i at which the present worth function reduces to zero



So, one has to start with an intuitive value of i and check whether the present worth function is positive. If so, increase the value of i until PW(i) becomes negative. Then, the rate of return is determined by interpolation method in the range of values of i for which the sign of the present worth function changes from positive to negative

EXAMPLE 13

Chapter 5: Replacement and Maintenance Analysis

Organizations providing goods/services use several facilities like equipment and machinery which are directly required in their operations. In addition to these facilities, there are several other items which are necessary to facilitate the functioning of organizations. All such facilities should be continuously monitored for their efficient functioning; otherwise, the quality of service will be poor. Besides the quality of service of the facilities, the cost of their operation and maintenance would increase with the passage of time. Hence, it is an absolute necessity to maintain the equipment in good operating conditions with economical cost. Thus, we need an integrated approach to minimize the cost of maintenance. In certain cases, the equipment will be obsolete over a period of time. If a firm wants to be in the same business competitively, it has to take decision on whether to replace the old equipment or to retain it by taking the cost of maintenance and operation into account. There are two basic reasons for considering the replacement of an equipment: **physical impairment of the various parts or obsolescence of the equipment**. Physical impairment refers only to changes in the physical condition of the machine itself. This would lead to a decline in the value of the service rendered, increased operating cost, increased maintenance cost or a combination of these. Obsolescence is due to improvement of the tools of production, mainly improvement in technology. So, it would be uneconomical to continue production with the same machine under any of the above situations. Hence, the machines are to be periodically replaced. Sometimes, the capacity of existing facilities may be inadequate to meet the current demand. Under such situation, the following alternatives will be considered.

- ✓ Replacement of the existing equipment with a new one.
- ✓ Augmenting the existing one with an additional equipment

5.1 Types of maintenance

Maintenance activity can be classified into two types: **preventive maintenance and breakdown maintenance**. **Preventive maintenance** (PM) is the periodical inspection and service activities which are aimed to detect potential failures and perform minor adjustments or repairs which will prevent major operating problems in future. **Breakdown maintenance** is the repair which is generally done after the equipment has attained down state. It is often of an emergency nature which will have associated penalty in terms of expediting cost of maintenance and down time cost of equipment. Preventive maintenance will reduce such cost up to a point. Beyond that point, the cost of preventive maintenance will be more when compared to the breakdown maintenance cost.

The total cost, which is the sum of the preventive maintenance cost and the breakdown maintenance cost, will go on decreasing with an increase in the level of maintenance up to a point. Beyond that point, the total cost will start increasing. The level of maintenance corresponding to the minimum total cost is the optimal level of maintenance. The concepts are demonstrated in Fig.

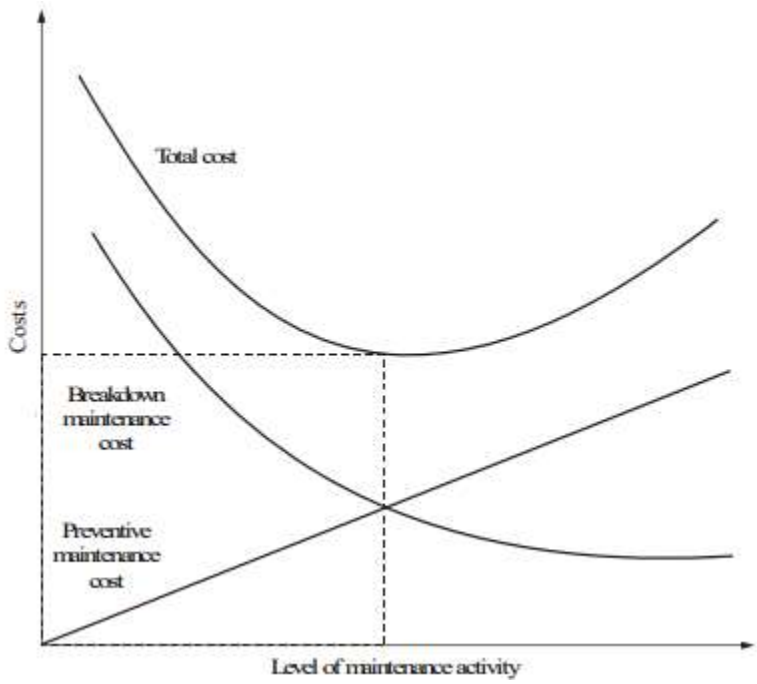


Fig. Maintenance costs

5.2 Types of replacement problem

Replacement study can be classified into two categories:

(a) Replacement of assets that deteriorate with time (Replacement due to gradual failure, or wear and tear of the components of the machines).

This can be further classified into the following types:

(i) Determination of economic life of an asset.

(ii) Replacement of an existing asset with a new asset.

(b) Simple probabilistic model for assets which fail completely (replacement due to sudden failure).

5.3 Determination of economic life of an asset

Any asset will have the following cost components:

- ✓ Capital recovery cost (average first cost), computed from the first cost (purchase price) of the machine.
- ✓ Average operating and maintenance cost (O & M cost)
- ✓ Total cost which is the sum of capital recovery cost (average first cost) and average maintenance cost.

A typical shape of each of the above costs with respect to life of the machine is shown in Fig

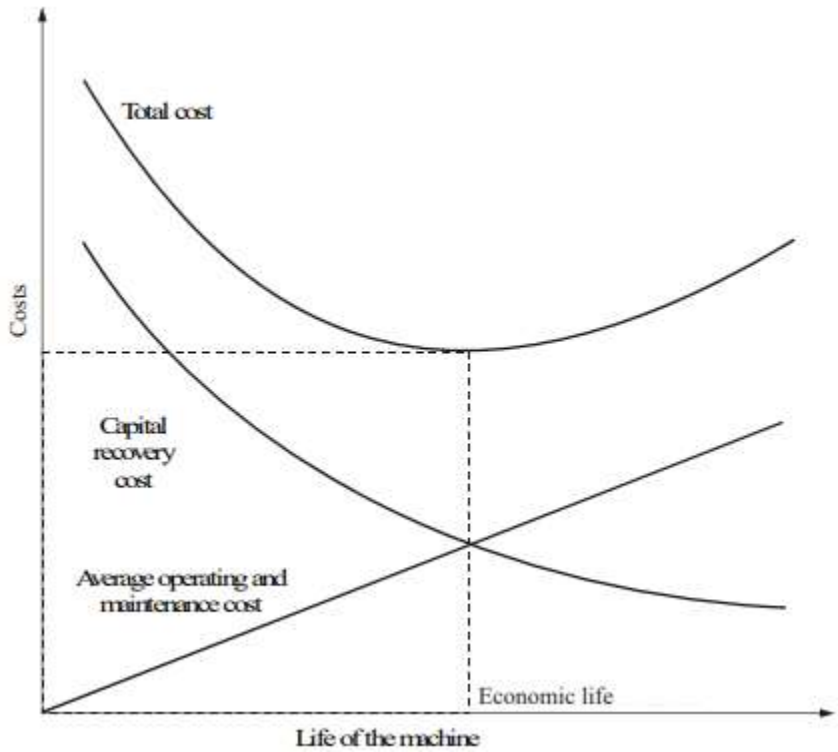


Chart showing economic life

From Fig., it is clear that the capital recovery cost (average first cost) goes on decreasing with the life of the machine and the average operating and maintenance cost goes on increasing with the life of the machine. From the beginning, the total cost continues to decrease up to a particular life and then it starts increasing. The point where the total cost is minimum is called the economic life of the machine. If the interest rate is more than zero per cent, then we use interest formulas to determine the economic life. The replacement alternatives can be evaluated based on the present worth criterion and annual equivalent criterion.

5.4 Replacement of existing asset with a new asset

In this section, the concept of comparison of replacement of an existing asset with a new asset is presented. In this analysis, the annual equivalent cost of each alternative should be computed first. Then the alternative which has the least cost should be selected as the best alternative. Before discussing details, some preliminary concepts which are essential for this type of replacement analysis are presented.

5.4.1 Capital Recovery with Return Consider the following data of a machine.

Let

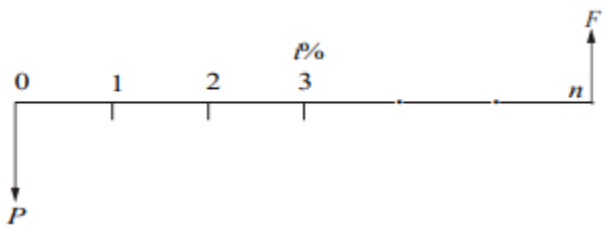
P = purchase price of the machine,

F = salvage value of the machine at the end of machine life,

n = life of the machine in years, and

i = interest rate, compounded annually

the corresponding cash flow diagram is shown in Fig.



Cash flow diagram of machine.

The equation for the annual equivalent amount for the above cash flow diagram is

$$AE(i) = (P - F) \times (A/P, i, n) + F \times i$$

This equation represents the capital recovery with return.

5.4.2 Concept of Challenger and Defender

If an existing equipment is considered for replacement with a new equipment, then the existing equipment is known as the defender and the new equipment is known as challenger. Assume that an equipment has been purchased about three years back for 5,00,000 frs and it is considered for replacement with a new equipment. The supplier of the new equipment will take the old one for some money, say, 3,00,000 frs. This should be treated as the present value of the existing equipment and it should be considered for all further economic analysis. The purchase value of the existing equipment before three years is now known as sunk cost, and it should not be considered for further analysis

5.5 Simple probabilistic model for items which fail completely

Electronic items like transistors, resistors, tube lights, bulbs, etc. could fail all of a sudden, instead of gradual deterioration. The failure of the item may result in complete breakdown of the system. The system may contain a collection of such items or just one item, say a tube light. Therefore, we use some replacement policy for such items which would avoid the possibility of a complete breakdown. The following are the replacement policies which are applicable for this situation.

(i) Individual replacement policy. Under this policy, an item is replaced immediately after its failure.

(ii) Group replacement policy. Under this policy, the following decision is made:

At what equal intervals are all the items to be replaced simultaneously with a provision to replace the items individually which fail during a fixed group replacement period?

There is a trade-off between the individual replacement policy and the group replacement policy. Hence, for a given problem, each of the replacement policies is evaluated and the most economical policy is selected for implementation. This is explained with two numerical problems.

EXAMPLE 14

Chapter 6: Depreciation

Any equipment which is purchased today will not work for ever. This may be due to wear and tear of the equipment or obsolescence of technology. Hence, it is to be replaced at the proper time for continuance of any business. The replacement of the equipment at the end of its life involves money. This must be internally generated from the earnings of the equipment. The recovery of money from the earnings of an equipment for its replacement purpose is called depreciation fund since we make an assumption that the value of the equipment decreases with the passage of time. Thus, the word "depreciation" means decrease in value of any physical asset with the passage of time

6.1 Methods of depreciation

There are several methods of accounting depreciation fund. These are as follows:

1. Straight line method of depreciation
2. Declining balance method of depreciation
3. Sum of the year's digits method of depreciation
4. Sinking-fund method of depreciation
5. Service output method of depreciation

6.1.1 Straight Line Method of Depreciation

In this method of depreciation, a fixed sum is charged as the depreciation amount throughout the lifetime of an asset such that the accumulated sum at the end of the life of the asset is exactly equal to the purchase value of the asset. Here, we make an important assumption that inflation is absent. Let

P = first cost of the asset,

F = salvage value of the asset

n = life of the asset,

Bt = book value of the asset at the end of the period t,

Dt = depreciation amount for the period t. T

he formulae for depreciation and book value are as follows:

$$D_t = (P - F)/n$$
$$B_t = B_{t-1} - D_t = P - t \times [(P - F)/n]$$

EXAMPLE 15

6.1.2 Declining Balance Method of Depreciation

In this method of depreciation, a constant percentage of the book value of the previous period of the asset will be charged as the depreciation amount for the current period. This approach is a more realistic approach, since the depreciation charge decreases with the life of the asset which matches with the earning potential of the asset. The book value at the end of the life of the asset may not be exactly equal to the salvage value of the asset. This is a major limitation of this approach.

Let

P = first cost of the asset,

F = salvage value of the asset,

n = life of the asset,

Bt = book value of the asset at the end of the period t,

K = a fixed percentage, and

Dt = depreciation amount at the end of the period t.

The formulae for depreciation and book value are as follows

$$D_t = K \times B_{t-1}$$

$$B_t = B_{t-1} - D_t = B_{t-1} - K \times B_{t-1}$$

$$= (1 - K) \times B_{t-1}$$

The formulae for depreciation and book value in terms of P are as follows:

$$D_t = K(1 - K)^{t-1} \times P$$
$$B_t = (1 - K)^t \times P$$

6.1.3 Sum-of-the-Years-Digits Method of Depreciation

In this method of depreciation also, it is assumed that the book value of the asset decreases at a decreasing rate. If the asset has a life of eight years, first the sum of the years is computed as

$$\text{Sum of the years} = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36 = n(n + 1)/2$$

The rate of depreciation charge for the first year is assumed as the highest and then it decreases. The rates of depreciation for the years 1–8, respectively are as follows: 8/36, 7/36, 6/36, 5/36, 4/36, 3/36, 2/36, and 1/36. For any year, the depreciation is calculated by multiplying the corresponding rate of depreciation with (P – F).

$$D_t = \text{Rate} \times (P - F)$$
$$B_t = B_{t-1} - D_t$$

The formulae for Dt and Bt for a specific year t are as follows:

$$D_t = \frac{n - t + 1}{n(n + 1)/2} (P - F)$$

$$B_t = (P - F) \frac{(n - t)}{n} \frac{(n - t + 1)}{(n + 1)} + F$$

6.1.4 Sinking Fund Method of Depreciation

In this method of depreciation, the book value decreases at increasing rates with respect to the life of the asset. Let

P = first cost of the asset,

F = salvage value of the asset,

n = life of the asset,

i = rate of return compounded annually, A

= the annual equivalent amount,

Bt = the book value of the asset at the end of the period t, and

Dt = the depreciation amount at the end of the period t.

The loss in value of the asset (P – F) is made available an the form of cumulative depreciation amount at the end of the life of the asset by setting up an equal depreciation amount (A) at the end of each period during the lifetime of the asset

A = (P – F) x [A/F, i, n]

The fixed sum depreciated at the end of every time period earns an interest at the rate of i% compounded annually, and hence the actual depreciation amount will be in the increasing manner with respect to the time period. A generalized formula for Dt is

Dt = (P – F) x(A/F, i, n) x (F/P, i, t – 1)

The formula to calculate the book value at the end of period t is

Bt = P – (P – F) (A/F, i, n) (F/A, i, t)

The above two formulae are very useful if we have to calculate Dt and Bt for any specific period. If we calculate Dt and Bt for all the periods, then the tabular approach would be better

6.1.5 Service Output Method of Depreciation

In some situations, it may not be realistic to compute depreciation based on time period. In such cases, the depreciation is computed based on service rendered by an asset.

Let

P = first cost of the asset

F = salvage value of the asset

X = maximum capacity of service of the asset during its lifetime

x = quantity of service rendered in a period.

Then, the depreciation is defined per unit of service rendered:

Depreciation/unit of service = (P – F)/X

Depreciation for x units of service in a period = $\frac{P-F}{X}(x)$

Chapter 7: Evaluation of Public Alternatives

In evaluating alternatives of private organizations, the criterion is to select the alternative with the maximum profit. The profit maximization is the main goal of private organizations while providing goods/services as per specifications to their customers. But the same criterion cannot be used while evaluating public alternatives. Examples of some public alternatives are constructing bridges, roads, dams, establishing public utilities, etc. The main objective of any public alternative is to provide goods/services to the public at the minimum cost. In this process, one should see whether the benefits of the public activity are at least equal to its costs. If yes, then the public activity can be undertaken for implementation. Otherwise, it can be cancelled. This is nothing but taking a decision based on Benefit-Cost ratio (BC) given by

$$\text{BC ratio} = \frac{\text{Equivalent benefits}}{\text{Equivalent costs}}$$

The benefits may occur at different time periods of the public activity. For the purpose of comparison, these are to be converted into a common time base (present worth or future worth or annual equivalent). Similarly, the costs consist of initial investment and yearly operation and maintenance cost. These are to be converted to a common time base as done in the equivalent benefits. Now the ratio between the equivalent benefits and equivalent costs is known as the "Benefit-Cost ratio". If this ratio is at least one, the public activity is justified; otherwise, it is not justified. Let

B_P = present worth of the total benefits

B_F = future worth of the total benefits

B_A = annual equivalent of the total benefits

P = initial investment

P_F = future worth of the initial investment

P_A = annual equivalent of the initial investment

C = yearly cost of operation and maintenance

C_P = present worth of yearly cost of operation and maintenance

C_F = future worth of yearly cost of operation and maintenance

$$\text{BC ratio} = \frac{B_P}{P + C_P} = \frac{B_F}{P_F + C_F} = \frac{B_A}{P_A + C}$$

EXAMPLE 16

CHAPTER 8: INFLATION ADJUSTED DECISIONS

A general inflationary trend in the cost of goods is common everywhere due to various interacting factors. If the rate of inflation is very high, it will produce extremely serious consequences for both individuals and institutions. Inflation is the rate of increase in the prices of goods per period. So, it has a compounding effect. Thus, prices that are inflated at a rate of 7% per year will increase 7% in the first year, and for the next year the expected increase will be 7% of these new prices. The same is true for succeeding years and hence the rate of inflation is compounded in the same manner that an interest rate is compounded. If the average inflation over six years period is 7%, then the prices at the beginning of the seventh year would be 150% that of the first year by assuming 100% for the prices at the beginning of the first year of the six-year period. If economic decisions are taken without considering the effect of inflation into account, most of them would become meaningless and as a result the organizations would end up with unpredictable return. But there is always difficulty in determining the rate of inflation. The world-wide trend/wish is to curtail inflation. But due to various reasons, it is very difficult to have zero inflation. For practical decision making, an average estimate may be assumed depending on the period of the proposals under consideration. Hence, we need a procedure which will combine the effects of inflation rate and interest rate to take realistic economic decision

8.1 PROCEDURE TO ADJUST INFLATION

A procedure to deal with this situation is summarized now.

1. Estimate all the costs/returns associated with an investment proposal in terms of today's value.
2. Modify the costs/returns estimated in step 1 using an assumed inflation rate so that at each future date they represent the costs/returns at that date in terms of the value that must be expended/received at that time, respectively.
3. As per our requirement, calculate either the annual equivalent amount or future amount or present amount of the cash flow resulting from step 2 by considering the time value of money.

EXAMPLE 17

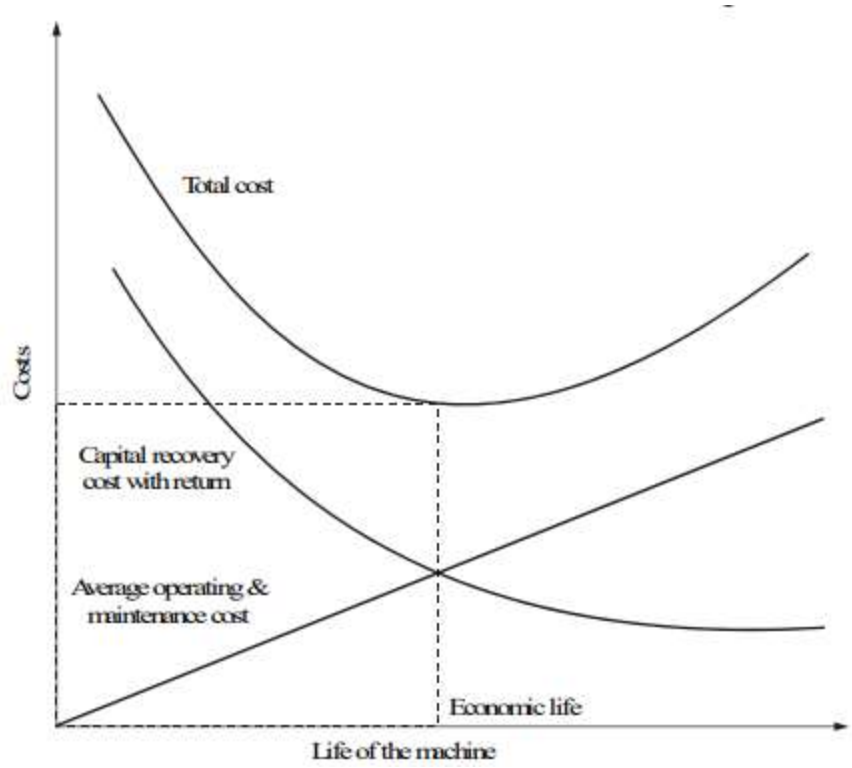
8.2 NFLATION ADJUSTED ECONOMIC LIFE OF MACHINE

In any industrial/service organization, equipment/machinery forms an important element. The productivity of any organization is a function of many factors. It is largely affected by efficient and effective use of machinery and equipment. So, operations and maintenance of these equipment are very important to the organization. A machine which is purchased today cannot be used forever. It has a definite economic lifetime. After the economic life, the machine should be replaced with a substitute machine with similar operational capabilities. This kind of analysis is called replacement analysis

The elements of costs involved in the replacement analysis are as follows:

1. Purchase cost (initial cost)
2. Annual operation and maintenance cost
3. Salvage value at the end of every year, if it is significant

The trade-off between different cost elements is shown in Fig



From Fig, it is clear that the sum of operation and maintenance cost increases with the life of the machine. But the capital recovery with return decreases with the life of the machine. The total cost of the machine goes on decreasing initially but it starts increasing after some years. The year with the minimum total cost is called as the economic life of the machine.

Limitation of Existing Model

In the case where the machine is replaced due to wear and tear, the following costs are considered

1. Initial cost
2. Operation and maintenance cost
3. Salvage value

In the existing model to deal with this type of replacement analysis, the different cost elements are estimated without taking the effect of inflation into account. The annual cost of operation and maintenance of the machine will increase with the age of the machine due to decline in efficiency of the machine. In the existing model, this increase in the operation and maintenance cost is taken into account. But the increase in the operation and maintenance cost due to inflation is not considered. Similarly, in the existing model, the salvage value is estimated without taking into account the effect of inflation. To highlight this particular fact on salvage value, an example is

now given. The internal combustion engines (R.A. Lister) which were made in England during pre-independence of India are still functioning well. Their resale value is going up year after year. This may be partly due to inflation and partly due to good quality of the engine parts. So, consideration of the effect of the inflation on the economic life of the machine is a realistic approach. In replacement analysis, a discount rate is usually assumed to reflect the time value of money. First the concept of replacement analysis is demonstrated without taking the inflation into account. Then, the same is demonstrated by taking the effect of inflation into account. At the end, a comparison between the two models is presented

EXAMPLE 18